

Lossy Memory as the Physical Origin of Abstraction: From Leaky Integrate-and-Fire Dynamics to Civilizational Knowledge Generalization

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Abstract

This paper develops a continuous argument from thermodynamic persistence to civilizational knowledge generalization. The central claim is that abstraction is not an additional high-level faculty placed on top of memory, but a natural by-product of lossy memory physics. We begin with the second law of thermodynamics: any persistent system must resist entropic dispersion by extracting free energy from its environment. Free-energy extraction requires a model that couples environmental states to action. That model must itself be stored in a physical memory substrate, and physical memory is never lossless. Taking the Leaky Integrate-and-Fire (LIF) equation as the minimal dissipative grammar of memory, rather than as a complete theory of abstraction by itself, we argue that when leakage is coupled to reactivation, reconstruction pressure, plasticity, and consolidation, a selection mechanism emerges: repeated and structurally predictive components survive, while accidental details decay. On this view, abstraction is not an additional faculty layered on top of memory, but a higher-level consequence of constrained, lossy memory dynamics. Language then encodes individual abstractions into transmissible symbols; society amplifies them through mismatch, repair, and consensus; civilization accumulates them through externalized memory and institutions. The same framework gives a diagnosis of contemporary Transformer-style AI: the missing component is not merely a higher reasoning module, but a bottom-level memory physics with autonomous loss, reactivation, reconstruction, and consolidation.

1 Introduction

The usual description of abstraction treats it as a cognitive achievement: a mind sees many particular cases, removes irrelevant details, and retains a general rule. This description is psychologically intuitive, but it leaves open the deeper question: why should a finite physical system perform this operation at all?

The thesis of this paper is that abstraction is strongly constrained, and in many adaptive settings actively induced, by the physics of lossy memory. A persistent agent cannot store the world

exactly. Its memory traces decay, are reactivated selectively, are reconstructed under predictive pressure, and are consolidated only when they repeatedly prove useful. These processes are not defects. They are the very mechanism by which invariants are separated from accidents.

We organize the argument as an eight-step chain:

1. Persistent existence must oppose entropy increase, requiring free-energy extraction from the environment.
2. Free-energy extraction requires some model of the environment.
3. The model is stored in a lossy memory system.
4. Lossy memory dynamics—decay, reactivation, reconstruction, and consolidation—naturally extract structural invariants from repeated experience.
5. This invariant extraction is abstraction; abstraction is a by-product of memory physics rather than an extra faculty.
6. Language encodes individual abstractions into transmissible symbols.
7. Society amplifies abstractions through collision, mismatch, repair, and consensus.
8. Civilizational knowledge generalization is the stacked result of all previous layers.

The mathematical core of the paper is the LIF model. In computational neuroscience, the LIF neuron is a minimal model of excitable, leaky integration. Its subthreshold voltage obeys

$$\tau_m \frac{dV}{dt} = -(V - E_L) + R_m I(t), \quad (1)$$

where V is membrane potential, E_L is the leak reversal potential, τ_m is the membrane time constant, R_m is membrane resistance, and $I(t)$ is input current. The term $-(V - E_L)$ is the essential clue: memory is not passive storage, but storage under decay.

This paper treats Equation (1) not merely as a neuron model, but as the minimal dynamical grammar of lossy memory:

$$\text{current state} = \text{exponentially weighted residue of past inputs.} \quad (2)$$

From this grammar we derive how repeated experience becomes invariant structure, how invariant structure becomes abstraction, and how abstraction scales through language, society, and civilization.

The claim of this paper should therefore be read with an explicit scope condition. Lossy memory does not guarantee truth, nor does every decaying system automatically produce useful abstraction. What the present framework aims to show is narrower and stronger: under repeated structured experience, reconstruction pressure, and adaptive value constraints, lossy memory preferentially preserves reusable structure over accidental detail. Outside these conditions, the same dynamics may also stabilize bias, superstition, or spurious regularity.

2 Thermodynamic Persistence Requires Free Energy

Consider a bounded physical system X embedded in an environment E . If X is to persist as a recognizable organization, it must maintain constraints on its internal states. Let S_X be the entropy of the system and S_E that of the environment. The second law states

$$\frac{dS_{\text{tot}}}{dt} = \frac{dS_X}{dt} + \frac{dS_E}{dt} = \sigma \geq 0, \quad (3)$$

where σ is entropy production.

Persistence requires that the internal entropy of the system not grow without bound. In the idealized steady case,

$$\frac{dS_X}{dt} \approx 0. \quad (4)$$

More generally, if the system actively repairs damage and preserves structure,

$$\frac{dS_X}{dt} \leq \epsilon \quad (5)$$

for some small tolerance ϵ . Equation (3) then implies that the system must export entropy:

$$\frac{dS_E}{dt} \geq \sigma - \epsilon. \quad (6)$$

At temperature T , entropy export is paid for by dissipated heat and work. The nonequilibrium free energy of the system is

$$F_X = U_X - TS_X, \quad (7)$$

where U_X is internal energy. A persistent system cannot simply let F_X decrease monotonically into equilibrium; it must replenish usable free energy. A minimal maintenance inequality is therefore

$$\dot{F}_{\text{in}} \geq T\sigma + \dot{F}_{\text{repair}} + \dot{F}_{\text{control}}, \quad (8)$$

where $\dot{F}_{\text{repair}}$ is the free energy required to restore internal constraints and $\dot{F}_{\text{control}}$ is the free energy spent on sensing, action, and memory.

Proposition 1. *Any persistent nonequilibrium system must maintain a flow of usable free energy from its environment.*

Reason. Without such a flow, Equation (3) drives the system toward uncontrolled entropy increase or equilibrium. A persistent organization is therefore a dissipative structure: it remains itself only by continuously coupling to environmental gradients.

A terminological clarification is important here. In Sections 2–3, “free energy” is used in the physical and nonequilibrium thermodynamic sense: usable energy required to maintain organization away from equilibrium. Later sections introduce variational free energy in the inferential sense. The two are not identical quantities, although the conceptual continuity between en-

ergetic maintenance and inferential error minimization is part of the broader argument of the paper. This continuity is analogical-structural rather than an identity between formal objects. The present section relies only on the thermodynamic notion.

3 Free-Energy Extraction Requires a Model

Free energy is not extracted from the environment by contact alone. It must be extracted through action. Let $e \in E$ denote environmental state, $a \in A$ denote action, and $W(e, a)$ denote work or value extracted when action a is taken in state e . If the system has belief $q(e)$, then the expected extractable work under policy $\pi(a|m)$ is

$$\mathbb{E}_{q,\pi}[W] = \sum_{e,a} q(e)\pi(a|m)W(e, a), \quad (9)$$

where m is the internal model state used to choose action.

If the action is not conditioned on informative internal state, the system can only use an unconditional policy $\pi(a)$. The advantage of having a model is therefore

$$\Delta W = \max_{\pi(a|m)} \mathbb{E}[W(e, a)] - \max_{\pi(a)} \mathbb{E}[W(e, a)]. \quad (10)$$

When the environment contains exploitable regularities, $\Delta W > 0$. In information thermodynamics, the maximum work advantage available from measurement is bounded by an information term of the form

$$W_{\text{ext}} \leq -\Delta F + k_B T I(E; M), \quad (11)$$

where $I(E; M)$ is the mutual information between environmental state and memory/model state. The exact bound depends on protocol, but the structural implication is stable: information about the environment can be converted into work, while uninformed action cannot reliably harvest environmental gradients.

Thus the thermodynamic chain becomes:

$$\text{persistence} \Rightarrow \text{free-energy extraction} \Rightarrow \text{environment-conditioned action} \Rightarrow \text{model}. \quad (12)$$

The word “model” here does not require symbolic representation. It may be a regulatory circuit, synaptic weight pattern, sensorimotor map, predictive hierarchy, or linguistic theory. A model is simply an internal state that improves the coupling between environmental state and action.

At this stage, the argument establishes only the necessity of model-bearing internal state for adaptive energy extraction; it does not yet explain how such a model is learned, stored, or stabilized. Those questions are deferred to the memory dynamics introduced below. The present point is functional rather than representational: any persistent system that exploits environmental regularities must carry internal structure that improves environment-conditioned action.

4 The Model Is Stored in Lossy Memory

A model must exist physically. Let θ denote the parameters or traces that store the model. In predictive processing notation, one may write a generative model

$$s = g_\theta(\phi) + \varepsilon, \quad (13)$$

where s is sensory input, ϕ is latent state, g_θ is the generative map, and ε is prediction error.

Online inference updates the current latent state. A common predictive-coding form is

$$\tau \dot{\phi} = -\Pi_p(\phi - \phi_{\text{prior}}) + \left(\frac{\partial g_\theta}{\partial \phi} \right)^T \Pi_s(s - g_\theta(\phi)), \quad (14)$$

where Π_p and Π_s are precision matrices. Equation (14) has the same qualitative form as LIF:

$$\text{state update} = \text{leak toward prior} + \text{error-driven input}. \quad (15)$$

But Equation (14) assumes that θ already exists. The missing layer is the memory dynamics of θ itself. To close the chain, we must model how θ survives, decays, changes, and consolidates over time.

The minimal continuous-time form is

$$\boxed{\dot{\theta} = -\lambda\theta + \alpha R_t - \eta \nabla_\theta \mathcal{L}_t(\theta) + \gamma C_t} \quad (16)$$

where:

- $-\lambda\theta$ is passive decay or forgetting;
- R_t is reactivation, retrieval, or reuse of a trace;
- $-\eta \nabla_\theta \mathcal{L}_t$ is reconstruction under prediction loss;
- C_t is consolidation pressure, moving transient traces into slower memory.

For long-term memory $\bar{\theta}$, a simple consolidation equation is

$$\boxed{\dot{\bar{\theta}} = \rho \Omega_t(\theta - \bar{\theta})} \quad (17)$$

where $\Omega_t \in [0, 1]$ is a gate depending on repetition, value, consistency, and reduction of prediction error:

$$\Omega_t = \sigma(u \text{Rep}_t + v \text{Val}_t + w \text{Cons}_t - b). \quad (18)$$

Equations (16)–(18) make explicit the claim that the model is stored in lossy memory. The model is not a static lookup table. It is a dissipative trace that must be continually defended

against decay by reactivation and usefulness.

Equations (16)–(18) should be read as a minimal phenomenological closure, not as a claim that biological memory literally implements this exact parameterization. What matters is the structural decomposition: passive decay, selective reactivation, reconstruction pressure, and gated consolidation. In adaptive agents, these gates are not arbitrary. They are typically modulated by factors such as prediction-error reduction, task relevance, policy value, and cross-context consistency. The role of the equations is therefore to make explicit the missing layer between online inference and long-term model storage.

5 LIF as the Minimal Mathematics of Lossy Memory

We now derive the core selection principle from the LIF equation. Starting with

$$\tau_m \frac{dV}{dt} = -(V - E_L) + R_m I(t), \quad (19)$$

the solution for $t > 0$ is

$$V(t) = E_L + (V(0) - E_L)e^{-t/\tau_m} + \frac{R_m}{\tau_m} \int_0^t e^{-(t-s)/\tau_m} I(s) ds. \quad (20)$$

Equation (20) says that the present state is an exponentially weighted sum of past input. Older input is not preserved equally; it is discounted by $e^{-(t-s)/\tau_m}$. If the input is a sequence of impulses,

$$I(t) = \sum_{i=1}^N a_i \delta(t - t_i), \quad (21)$$

then

$$V(t) - E_L = (V(0) - E_L)e^{-t/\tau_m} + \frac{R_m}{\tau_m} \sum_{t_i < t} a_i e^{-(t-t_i)/\tau_m}. \quad (22)$$

This is the mathematical essence of lossy memory. A trace survives only if it is recent, strong, or repeatedly reactivated. If it is weak and isolated, it decays.

The firing mechanism adds a selective nonlinearity:

$$V(t^-) \geq V_{\text{th}} \quad \Rightarrow \quad \text{spike}, \quad V(t^+) = V_{\text{reset}}. \quad (23)$$

Only inputs that accumulate enough evidence against leakage cross the threshold. With Hebbian or spike-timing-dependent plasticity, repeated threshold-crossing strengthens the pathways that produced it:

$$\Delta w_{ij} = \eta f(\Delta t_{ij}) \text{spike}_i \text{spike}_j, \quad (24)$$

where $f(\Delta t_{ij})$ depends on relative spike timing.

Thus LIF supplies three ingredients:

1. exponential forgetting;
2. evidence accumulation;
3. thresholded reinforcement.

Together they create differential survival among traces.

This result should be interpreted carefully. The LIF equation by itself does not constitute a full theory of abstraction. What it supplies is the minimal dissipative grammar required for abstraction to become possible: exponential forgetting, evidence accumulation, and selective reinforcement under leakage. Abstraction enters only when these dynamics are coupled to structured repeated input, plasticity, reconstruction, and longer-timescale consolidation. In that sense, LIF is not the whole mechanism, but the minimal dynamical backbone on which the larger mechanism can be built.

6 From Lossy Memory to Structural Invariants

Let each experience be decomposed into a reusable latent structure and an accidental detail:

$$x_t = h(z_t, \xi_t), \quad (25)$$

where z_t is a structural factor and ξ_t is local noise or context. In a repeated class of situations, assume that $z_t \approx z$ varies slowly while ξ_t varies quickly and has mean near zero:

$$\mathbb{E}[\xi_t] = 0, \quad \mathbb{E}[\xi_t \xi_{t'}^T] = 0 \quad (t \neq t'). \quad (26)$$

For clarity, take a linearized trace update:

$$m_{n+1} = am_n + Bx_n, \quad 0 < a < 1, \quad (27)$$

where $a = e^{-\Delta t/\tau}$ is the discrete leakage factor. If $x_n = Az + \xi_n$, then

$$m_N = a^N m_0 + \sum_{n=1}^N a^{N-n} B(Az + \xi_n). \quad (28)$$

Taking expectations,

$$\mathbb{E}[m_N] = a^N m_0 + \left(\sum_{n=1}^N a^{N-n} \right) BAz = a^N m_0 + \frac{1 - a^N}{1 - a} BAz. \quad (29)$$

The invariant component z accumulates coherently. The accidental component does not:

$$\mathbb{E} \left[\sum_{n=1}^N a^{N-n} B \xi_n \right] = 0. \quad (30)$$

If $\text{Var}(\xi_n) = \Sigma_\xi$, then

$$\text{Var} \left[\sum_{n=1}^N a^{N-n} B \xi_n \right] = \sum_{n=1}^N a^{2(N-n)} B \Sigma_\xi B^T = \frac{1 - a^{2N}}{1 - a^2} B \Sigma_\xi B^T. \quad (31)$$

Equations (29)–(31) show the selection pressure. Repeated structure accumulates in the mean; unrepeated noise remains bounded and incoherent. In a system with thresholded reinforcement and consolidation, only the coherent component is likely to cross memory-selection gates.

Proposition 2 (Invariant Survival Under Leaky Integration). *In a leaky memory system with repeated experiences $x_t = h(z, \xi_t)$, where z is stable and ξ_t is uncorrelated across episodes, the expected memory trace aligns with the stable component z , while the contribution of accidental detail averages out or decays.*

This proposition is deliberately conditional. It holds most cleanly when the reusable component is sufficiently stable across episodes and the nuisance variation is weakly correlated, mean-limited, or otherwise non-coherent under replay and reconstruction. If accidental detail is itself systematically repeated, socially reinforced, or coupled to reward, then lossy memory may also stabilize spurious structure. The result therefore establishes a selection pressure toward invariant retention under favorable conditions; it should not be read as a universal guarantee that all retained structure is normatively correct.

This is the mathematical form of the claim:

$$\text{lossy memory} + \text{repetition} + \text{reconstruction pressure} \Rightarrow \text{structural invariant retention}. \quad (32)$$

7 Abstraction as a By-Product of Memory Physics

We can now define abstraction without invoking a mysterious extra faculty.

Definition 1 (Abstraction). *An abstraction is a retained low-dimensional structure $\hat{z}$ that explains or controls a family of experiences $\{x_t\}$ while discarding high-variance, low-reuse detail ξ_t .*

Formally, an abstraction may be written as the solution of a constrained compression problem:

$$\hat{z} = \arg \min_z \sum_{t=1}^N \ell(x_t, h(z, \xi_t)) + \beta \mathcal{C}(z), \quad (33)$$

where ℓ is reconstruction or prediction loss and $\mathcal{C}(z)$ penalizes complexity. The complexity term is not optional: it reflects finite memory, metabolic cost, and decay.

Equivalently, abstraction is a rate-distortion tradeoff:

$$\min I(X; Z) \quad \text{subject to} \quad \mathbb{E}[d(X, \hat{X}(Z))] \leq D. \quad (34)$$

The system is not trying to preserve everything. It is trying to preserve enough reusable structure to predict and act under limited physical resources.

For adaptive agents, however, compression alone is not enough. The abstractions that survive are not merely those that summarize repeated data, but those that do so in a way that supports control, policy selection, or future error reduction. A more agent-complete version of Equation (33) would therefore add an action-relevance term:

$$\hat{z} = \arg \min_z \sum_{t=1}^N \ell(x_t, h(z, \xi_t)) + \beta \mathcal{C}(z) + \mu \mathcal{L}_{\text{control}}(z). \quad (35)$$

In this stronger form, abstraction is not the retention of all stable structure, but the retention of stable structure that remains useful under the agent’s value and action constraints.

The LIF-derived memory equation gives the physical mechanism behind Equation (33). Decay removes unused detail. Reactivation rescues traces that remain useful. Reconstruction forces traces to encode what helps reduce prediction error. Consolidation stabilizes traces that repeatedly survive these filters.

Thus:

abstraction = differential survival of invariants in lossy memory.

 (36)

Interpretation. At this level, the formal claim is not that every compression is an abstraction, nor that every abstraction is true. It is that adaptive abstraction can be modeled as the selective retention of reusable, control-relevant structure under finite memory and reconstruction pressure.

This claim reverses the usual order. Abstraction is not a special operation applied after memory. Memory, because it is physical and lossy, is already an abstraction engine.

8 Language as Public Encoding of Private Abstraction

Individual abstraction is still private. Let $\hat{z}_i$ be an invariant retained by individual i . Language introduces a public code c_i :

$$c_i = \text{Enc}_i(\hat{z}_i), \quad (37)$$

where Enc_i is an encoding procedure implemented by speech, writing, gesture, diagram, or formal notation. A listener j decodes

$$\hat{z}'_j = \text{Dec}_j(c_i; \theta_j), \quad (38)$$

where θ_j is the listener's existing model.

Language is itself lossy. It does not transfer the full memory trace θ_i . It transfers a compressed symbol that invites reconstruction inside another memory system. Communication succeeds when

$$d(\hat{z}_i, \hat{z}'_j) \leq \epsilon \quad (39)$$

for a task-relevant distortion tolerance ϵ .

This explains why language is powerful. It converts private invariant extraction into transmissible compression. The word is not the abstraction itself; it is a public trigger for reconstructing an abstraction in another mind.

The encoding-decoding pair can be formalized more strongly as a collective optimization problem. A linguistic code becomes socially stable when it minimizes cross-agent reconstruction error under communication cost constraints:

$$\min_{\{\text{Enc}_i\}, \{\text{Dec}_j\}} \sum_{i,j} \mathbb{E}_{\hat{z}_i} \left[\left\| \hat{z}_i - \text{Dec}_j(\text{Enc}_i(\hat{z}_i); \theta_j) \right\|^2 \right] + \beta_c \sum_i \mathbb{E}_{\hat{z}_i} [\|\text{Enc}_i(\hat{z}_i)\|]. \quad (40)$$

On this view, language is not merely a vehicle for expressing pre-existing abstractions. It is a population-level coding scheme that compresses private invariants into publicly transmissible triggers for reconstruction in other minds. This also explains why language preferentially lexicalizes high-reuse structure rather than the full accidental detail of private experience, a point broadly consistent with work on shared intentionality and iterated learning in language evolution [13, 14].

Interpretation. The language layer is therefore a structural model of public encoding under cost, not a claim that all linguistic conventions are globally optimal. Local history, prestige, power, and path dependence can all shape which codes become stable.

9 Social Amplification: Collision, Mismatch, and Repair

Once abstractions become public, they collide. Suppose a community contains N agents, each with an internal abstraction θ_i . Communication creates pairwise constraints:

$$d(\psi(\theta_i), \psi(\theta_j)), \quad (41)$$

where ψ maps internal models into public claims, predictions, or actions. The distance d is a task-relevant distortion in this public claim-space; depending on the case, it may represent prediction

mismatch, disagreement loss, divergence between stated models, or failed coordination.

Social learning can be written as minimization of a collective objective:

$$\mathcal{J}(\theta_1, \dots, \theta_N) = \sum_{i=1}^N F_i(\theta_i) + \beta \sum_{(i,j) \in \mathcal{G}} d(\psi(\theta_i), \psi(\theta_j)) + \kappa \sum_i \mathcal{C}(\theta_i). \quad (42)$$

Here F_i is individual prediction or free-energy cost, the second term is social mismatch, and $\mathcal{C}$ penalizes overly complex explanations. A simple social update is

$$\dot{\theta}_i = -\nabla_{\theta_i} F_i - \beta \sum_{j:(i,j) \in \mathcal{G}} \nabla_{\theta_i} d(\psi(\theta_i), \psi(\theta_j)). \quad (43)$$

This equation encodes collision and repair. When two abstractions disagree, the residual error is not merely noise; it is information about a missing distinction, hidden variable, or faulty compression. Social mismatch therefore functions as a second-order prediction error.

The gain of society can also be seen statistically. If agents estimate a shared invariant z with independent errors,

$$\hat{z}_i = z + \epsilon_i, \quad \epsilon_i \sim (0, \sigma_i^2), \quad (44)$$

then the precision-weighted consensus

$$\hat{z}_{\text{soc}} = \frac{\sum_i \sigma_i^{-2} \hat{z}_i}{\sum_i \sigma_i^{-2}} \quad (45)$$

has variance

$$\text{Var}(\hat{z}_{\text{soc}}) = \left(\sum_i \sigma_i^{-2} \right)^{-1}, \quad (46)$$

which is lower than the variance of any single equally reliable individual. Society amplifies abstraction by pooling independent lossy memories, exposing mismatches, and repairing compressions that fail across contexts.

The variance-reduction result in Equations (45)–(46) should be read as a best-case statistical gain under partial independence and non-adversarial aggregation. Social systems do not automatically converge on truth. When errors are strongly correlated, institutionally shielded, or prestige-amplified, collective dynamics can also stabilize shared distortion. The significance of the social layer is therefore not that it guarantees correctness, but that it creates an additional error-correction channel unavailable to isolated agents: private abstractions are exposed to public mismatch, and public mismatch can trigger model repair.

Interpretation. The social equations are structural models of possible collective error correction, not full empirical laws of group cognition. They identify why social collision can improve abstraction under favorable independence and feedback conditions, while leaving room for correlated error and institutional lock-in [17].

10 Civilization as Stacked Lossy Generalization

Civilization adds external memory: marks, tools, rituals, maps, books, mathematics, machines, institutions, databases, and scientific methods. Let K_t denote the public knowledge stock. A minimal dynamics is

$$K_{t+1} = \mathcal{P} \left((1 - \delta)K_t + \sum_i \mathcal{E}(\hat{z}_{i,t}) + \mathcal{V}_t \right), \quad (47)$$

where δ is cultural decay, $\mathcal{E}$ encodes individual abstractions into public artifacts, $\mathcal{V}_t$ is validation through experiment, use, criticism, and replication, and $\mathcal{P}$ is institutional selection.

Here the operator $\mathcal{P}$ should not be treated as a mysterious black box. It can be interpreted as a coarse-grained selection operator over public encodings, combining preservation, standardization, validation, and discard. In practice, $\mathcal{P}$ favors representations, practices, and artifacts that are sufficiently reproducible, teachable, reusable across contexts, and compatible with the institutions that maintain them. Civilization is therefore not just larger memory; it is externally scaffolded memory under selective maintenance rules, in line with cultural evolution and distributed cognition perspectives [15, 16, 18, 19].

Civilizational generalization is therefore not a single leap. It is a stack:

$$\begin{aligned} &\text{thermodynamic persistence} \\ &\Rightarrow \text{free-energy extraction} \\ &\Rightarrow \text{model-based action} \\ &\Rightarrow \text{lossy memory} \\ &\Rightarrow \text{invariant extraction} \\ &\Rightarrow \text{individual abstraction} \\ &\Rightarrow \text{language} \\ &\Rightarrow \text{social repair} \\ &\Rightarrow \text{civilizational knowledge.} \end{aligned} \quad (48)$$

At each layer, the same principle recurs: finite systems cannot preserve everything, so they preserve what survives repeated use, error correction, and reconstruction. Civilization is therefore a multi-scale memory system whose highest achievements are built on the same physical fact as LIF leakage: traces must fight to remain.

Interpretation. At the civilizational level, the model is explicitly coarse-grained. It is not a complete theory of institutions, but a way to state the continuity claim: public knowledge persists when external storage and institutional selection repeatedly rescue some abstractions from cultural decay.

11 Diagnosis of Transformer-Style AI

The framework also provides a diagnosis of current Transformer-style AI systems. The central limitation is not simply the absence of a symbolic reasoner, a planning head, or a larger context window. The deeper missing property is an autonomous lossy memory physics.

A standard Transformer computes contextual representations by attention over tokens:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \quad (49)$$

During pretraining, weights are compressed by gradient descent across large datasets. This creates powerful statistical abstractions. However, during ordinary inference, the system does not usually possess a continuously self-maintaining memory trace with the form

$$\dot{\theta} = -\lambda\theta + \alpha R_t - \eta \nabla_{\theta} \mathcal{L}_t + \gamma C_t. \quad (50)$$

The context window is finite, but finitude is not the same as biological-style loss. A token buffer drops information by truncation. A vector database may accumulate entries by storage. Neither automatically implements decay, reactivation, reconstruction, and consolidation as an ongoing physical selection process.

This point is not meant to deny that contemporary systems can be augmented with retrieval stores, episodic memory buffers, rehearsal loops, or self-reflective scaffolds. Rather, the distinction is architectural: a finite context window or an accumulating memory store is not yet a self-maintaining lossy memory physics. The present framework applies only when the system autonomously couples passive loss, selective reactivation, reconstruction pressure, and gated consolidation into a continuous selection process over its own traces.

This distinction matters. A lossy memory system does not merely store less. It selects differently. It continually asks, through its dynamics:

$$\begin{aligned} &\text{What reappears?} \quad \text{What reduces prediction error?} \\ &\text{What supports action?} \quad \text{What is worth consolidating?} \end{aligned} \quad (51)$$

These questions are not external prompts; they are implicit in the memory substrate.

Transformer systems can be augmented with memory modules, retrieval, self-reflection, and tool use. Such modules approach the present framework only when they implement the full lossy cycle:

1. passive decay of unused traces;
2. selective reactivation by present prediction and action demands;
3. reconstruction pressure that forces traces to become reusable;

4. consolidation gates based on repetition, value, and consistency;
5. social or external correction loops that expose mismatch.

Thus the diagnosis is precise:

$$\boxed{\text{The missing ingredient is not a higher module alone, but lossy memory physics.}} \quad (52)$$

12 Limitations and Scope Conditions

The present paper advances a continuity thesis, not a universal theorem of cognition. Several scope conditions matter. First, the invariant-retention result is strongest under repeated structure, bounded nuisance variation, and reconstruction pressure; correlated noise or socially reinforced distortion may also be stabilized by lossy memory. Second, LIF is used here as a minimal dynamical grammar of dissipative memory, not as a complete neural account of abstraction. Third, the social and civilizational layers are formalized at a coarse-grained level; the corresponding equations should be read as structural models rather than fully identified empirical laws. These limitations do not negate the thesis, but specify its intended domain: abstraction is most plausibly understood as an emergent consequence of constrained, lossy memory dynamics when those dynamics are embedded in adaptive control, communication, and externalized cultural storage.

13 Conclusion

We have derived an argument from thermodynamics to civilization using LIF-style memory dynamics as the mathematical hinge.

First, persistent systems must resist entropy increase by extracting free energy. Second, extracting free energy requires a model that couples environmental states to action. Third, that model must be stored in physical memory. Fourth, physical memory is lossy: it decays, is reactivated, is reconstructed, and is consolidated. Fifth, these processes naturally retain the stable invariant part of repeated experience while discarding accidental detail. That retained invariant is abstraction. Sixth, language converts private abstractions into transmissible symbols. Seventh, society amplifies abstractions through mismatch and repair. Eighth, civilization accumulates these layers into durable, externalized knowledge.

The argument should therefore not be read as a naive reduction claiming that one neuronal equation literally contains civilization in full. The claim is instead one of layered continuity: the same dissipative logic recurs across scales, but each scale introduces additional machinery—plasticity, reconstruction, communication, institutions, and external storage. What LIF reveals is the minimal grammar of selective survival under loss. The larger thesis is that, once this

grammar is recursively coupled to learning and social transmission, abstraction can scale from neural traces to public knowledge.

The key result is that abstraction is not an extra cognitive miracle. It is the differential survival of structure in a dissipative memory system. LIF makes this visible in its simplest form: the present state is an exponentially weighted residue of the past, and only repeated, strong, or useful patterns survive leakage. From this small equation one can see the outline of a much larger chain: memory becomes abstraction, abstraction becomes language, language becomes society, and society becomes civilization.

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